Bracketing hyperoperations: a rank dichotomy for the Tamari order

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Abstract

Let H_r denote the Goodstein hyperoperation hierarchy, so that H_1 is addition, H_2 multiplication, H_3 exponentiation and H_4 tetration. Since H_r is non-associative for $r \geq 3$, the value of a bracketed expression $\mathrm{H}_r(a, \ldots, a)$ depends on the bracketing, i.e. on a binary tree. We study how the value varies over the Tamari lattice of such trees.

We prove that a *semi-associative law* runs through the whole hierarchy: for every rank $r \geq 3$ and all $x, y \geq 2$,

$$\mathrm{H}_r(\mathrm{H}_r(x, y), z) \;\leq\; \mathrm{H}_r(x, \mathrm{H}_r(y, z)),$$

so a right rotation at any node never decreases the value and evaluation is order-preserving from the Tamari lattice to $(\mathbb{N}, \leq)$; the left comb is a minimiser and the right comb, whose value is $\mathrm{H}_{r+1}(a, n)$, a maximiser.

Our main result is that the *degeneracy* of this law disappears at rank 4. We classify equality completely. For $x, y \geq 2$:

$$r = 3: \quad \text{equality} \iff z = 1 \ \text{ or } \ (y, z) = (2, 2); \qquad r \geq 4: \quad \text{equality} \iff z = 1.$$

Thus rank 3 carries exactly one non-trivial collision — already visible for $n = 4$ leaves, where the Tamari cover $((aa)a)a \lessdot (aa)(aa)$ has equal endpoints — and from rank 4 on the evaluation is strictly increasing along every such cover. The mechanism is identified: the collapsing law $(x^y)^z = x^{yz}$, which produces the rank-3 equality, has no analogue at rank 4 or above.

The result is not an asymptotic triviality. The two sides compare a growing *first* argument against a growing *second* one, and small arguments do not blow up with the rank: $\mathrm{H}_r(2, 2) = 4$ for every $r \geq 2$. Moreover the hypothesis $y \geq 2$ is sharp at every rank: for $y = 1$ the inequality reverses.

All statements are formalized and machine-checked in Lean 4; the development contains no `sorry` and every theorem is audited with `#print axioms`.

1 Introduction

Exponentiation is not associative, and it is classical that among all bracketings of $a^{a^{\cdot^{\cdot^{\cdot^{a}}}}}$ with n occurrences of $a \geq 2$ the right-associated one is the largest; its value is the tetration $a \uparrow\uparrow n$. What appears not to have been examined is the *order* structure behind this fact, and its behaviour further up the hyperoperation hierarchy.

Two strands of existing work bracket the question. On the one hand, the *associative spectrum* of a binary operation, introduced by Csákány and Waldhauser [1], measures non-associativity by counting how many distinct values the C_{n-1} bracketings can take; they prove in particular that exponentiation is *Catalan*, i.e. distinct bracketings induce distinct operations. That line of work counts values. On the other hand, the set of bracketings carries the Tamari lattice [7], generated by right rotations, whose proof-theoretic content was isolated by Zeilberger [8] as a semi-associative law. That line of work orders bracketings. The present paper connects the two: we show that for hyperoperations the Tamari order is *respected* by evaluation, and we locate precisely where the correspondence degenerates.

Results

Throughout, H_r is the hyperoperation hierarchy fixed in Section 2, with $\mathrm{H}_1(a,b) = a+b$, $\mathrm{H}_2(a,b) = ab$, $\mathrm{H}_3(a,b) = a^b$ and $\mathrm{H}_4(a,b) = a \uparrow\uparrow b$.

Theorem 1.1 (Rank 3, complete solution)**.** *Let $x \geq 1$ and $y \geq 2$. Then $(x^y)^z \leq x^{(y^z)}$ for every $z \geq 0$. If moreover $x \geq 2$, equality holds if and only if $z = 1$ or $(y,z) = (2,2)$. The hypothesis $y \geq 2$ cannot be dropped: for $y = 1$ the inequality fails.*

Theorem 1.2 (Rank 4)**.** *Let $x \geq 2$ and $y \geq 2$. Then $(x \uparrow\uparrow y) \uparrow\uparrow z \leq x \uparrow\uparrow (y \uparrow\uparrow z)$ for every $z \geq 0$. If moreover $z \geq 2$, the inequality is strict.*

Proposition 1.3 (Tamari monotonicity)**.** *Let $a \geq 2$ and $r \geq 3$. If $T \lessdot U$ is a Tamari cover, i.e. U arises from T by a right rotation at some node, then $\mathrm{ev}_{r,a}(T) \leq \mathrm{ev}_{r,a}(U)$. Consequently $\mathrm{ev}_{r,a}$ is order-preserving for the Tamari order. Moreover, for the right comb R_n with n leaves, $\mathrm{ev}_{r,a}(R_n) = \mathrm{H}_{r+1}(a,n)$.*

Corollary 1.4 (Dichotomy)**.** *The monotonicity of Proposition 1.3 is* not strict *at rank 3: the cover $((aa)a)a \lessdot (aa)(aa)$ carries equal values, both 256, when $a = 2$. For every $r \geq 4$, by contrast, a root rotation is strict as soon as the value of the third subtree is at least 2, which holds automatically for $a \geq 2$.*

Theorem 1.5 (All ranks)**.** *For every $r \geq 3$ and all $x \geq 2$, $y \geq 2$ and $z \geq 0$,*

$$\mathrm{H}_r\big(\mathrm{H}_r(x,y), z\big) \ \leq \ \mathrm{H}_r\big(x, \mathrm{H}_r(y,z)\big).$$

Theorem 1.6 (Complete equality classification)**.** *Let $x \geq 2$ and $y \geq 2$.*

1. *For every $r \geq 2$, equality holds in Theorem 1.5 when $z = 1$.*
2. *For $r = 3$ the only further equality is $(y,z) = (2,2)$.*
3. *For $r \geq 4$ and $z \geq 2$ the inequality is* strict.

Hence for $r \geq 4$ equality holds if and only if $z = 1$.

Theorems 1.5 and 1.6 are proved in Section 6 by inductions on the rank whose engine is the identity-like inequality (SUM_r) below; the case $r = 3$ is Theorem 1.1 and the case $r = 4$ can also be obtained directly, as in Section 4.

Remark 1.7 (The statement is not an asymptotic triviality)**.** Two features are worth isolating. First, (HT_r) does not follow from monotonicity alone: on the left the *first* argument grows, $x \mapsto \mathrm{H}_r(x,y) > x$, while on the right the *second* argument grows, $z \mapsto \mathrm{H}_r(y,z)$, so the two coordinates compete. Second, high rank does not make everything large: since $\mathrm{H}_r(a,1) = a$ for $r \geq 2$, one gets

$$\mathrm{H}_r(2,2) = \mathrm{H}_{r-1}(2, \mathrm{H}_r(2,1)) = \mathrm{H}_{r-1}(2,2) = \cdots = \mathrm{H}_2(2,2) = 4$$

for *every* $r \geq 2$. A degenerate corner therefore survives at every rank, and it is exactly this corner that carries the rank-3 equality of Theorem 1.6(2) and that has to be treated separately in the proof of Theorem 1.6(3).

Formalization

Every theorem stated here is proved in Lean 4 on top of Mathlib. The development defines the hyperoperation by the usual double recursion and *verifies the convention* by proving $\mathrm{H}_1 = +$, $\mathrm{H}_2 = \cdot$, $\mathrm{H}_3 = (\ )^{(\ )}$; this matters because the literature on hyperoperations uses several incompatible indexings and initial values. Section 7 describes the development.

2 Preliminaries

Definition 2.1 (Hyperoperation [3])**.** Define $\mathrm{H} : \mathbb{N} \times \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ by

$$\mathrm{H}_0(a, b) = b + 1, \quad \mathrm{H}_1(a, 0) = a, \quad \mathrm{H}_2(a, 0) = 0, \quad \mathrm{H}_{r+3}(a, 0) = 1,$$

$$\mathrm{H}_{r+1}(a, b + 1) = \mathrm{H}_r\big(a, \mathrm{H}_{r+1}(a, b)\big).$$

Lemma 2.2. *For all a, b we have $\mathrm{H}_1(a, b) = a+b$, $\mathrm{H}_2(a, b) = ab$ and $\mathrm{H}_3(a, b) = a^b$. Moreover $\mathrm{H}_r(a, 1) = a$ for $r \geq 2$.*

We write $a \uparrow\uparrow b := \mathrm{H}_4(a, b)$, so $a \uparrow\uparrow 0 = 1$ and $a \uparrow\uparrow (b + 1) = a^{(a\uparrow\uparrow b)}$.

Definition 2.3 (Bracketings and evaluation)**.** Let $\mathcal{T}_n$ denote the set of binary trees with n leaves. For $r, a \in \mathbb{N}$ define $\mathrm{ev}_{r,a} : \mathcal{T}_n \to \mathbb{N}$ by $\mathrm{ev}_{r,a}(\text{leaf}) = a$ and $\mathrm{ev}_{r,a}(T \cdot U) = \mathrm{H}_r\big(\mathrm{ev}_{r,a}(T), \mathrm{ev}_{r,a}(U)\big)$.

Definition 2.4 (Right rotation)**.** The *right rotation at the root* sends $((A \cdot B) \cdot C)$ to $(A \cdot (B \cdot C))$. The Tamari order on $\mathcal{T}_n$ is generated by right rotations performed at arbitrary nodes; that this order is a lattice, with the left comb as least and the right comb as greatest element, is the theorem of Friedman and Tamari [2], whose title already names the relation a *loi demi-associative* [7, 8].

Thus the inequality

$$\mathrm{H}_r\big(\mathrm{H}_r(x, y), z\big) \;\leq\; \mathrm{H}_r\big(x, \mathrm{H}_r(y, z)\big) \tag{HT$_r$}$$

is exactly the assertion that a right rotation, applied with x, y, z the values of the three subtrees, does not decrease the value.

3 Rank three

At rank 3 the collapsing law $(x^y)^z = x^{yz}$ reduces (HT$_r$) to a comparison of exponents.

Lemma 3.1. *If $y \geq 2$ then $yz \leq y^z$ for every z. If moreover $z \geq 3$ the inequality is strict.*

Proposition 3.2. *If $y \geq 2$ then $yz = y^z$ if and only if $z = 1$ or $(y, z) = (2, 2)$.*

Proof. If $z = 0$ then $yz = 0 \neq 1 = y^z$. If $z = 1$ both sides are y. If $z = 2$ then $yz = y^z$ reads $2y = y^2$, i.e. $y = 2$ after cancelling $y > 0$. If $z \geq 3$ the inequality is strict by Lemma 3.1. The converse is immediate. $\square$

Proof of Theorem 1.1. Since $(x^y)^z = x^{yz}$, the inequality follows from Lemma 3.1 and monotonicity of $t \mapsto x^t$ for $x \geq 1$. For $x \geq 2$ the map $t \mapsto x^t$ is injective, so equality holds iff $yz = y^z$, and Proposition 3.2 applies. For the last claim, $(x^1)^z = x^z$ while $x^{(1^z)} = x$, and $x^z > x$ whenever $x, z \geq 2$. $\square$

Example 3.3 (Failure of strictness)**.** Take $a = 2$ and $n = 4$. The five bracketings evaluate under H_3 to

$$((aa)a)a = 256, \quad (a(aa))a = 256, \quad (aa)(aa) = 256, \quad a((aa)a) = 65536, \quad a(a(aa)) = 65536.$$

In particular the Tamari cover $((aa)a)a \lessdot (aa)(aa)$ has equal endpoints, so $\mathrm{ev}_{3,2}$ is monotone but not strictly monotone; it is also not injective.

4 Rank four

At rank 4 no collapsing law is available and a different argument is required. The proof below is by induction on z; the base case $z = 1$ must be treated separately, since the inductive hypothesis at $z = 0$ is too weak to survive the step.

Lemma 4.1 (Squaring)**.** *If $x \geq 2$ and $m \geq 1$ then $(x \uparrow\uparrow m)^2 \leq x \uparrow\uparrow (m+1)$.*

Proof. Induction on m. For $m = 1$ the claim is $x^2 \leq x^x$, which holds as $x \geq 2$. For the step put $t = x \uparrow\uparrow m$; then $(x \uparrow\uparrow (m+1))^2 = (x^t)^2 = x^{2t} \leq x^{(x^t)} = x \uparrow\uparrow (m+2)$, where $2t \leq 2^t \leq x^t$ uses $t \geq 1$ and $x \geq 2$. □

Remark 4.2. The direct inequality $t^2 \leq x^t$ is *false* in general — take $x = 2$, $t = 3$ — so the inductive formulation in Lemma 4.1 is essential.

Lemma 4.3. *If $y \geq 2$ and $m \geq 2$ then $m + 2 \leq y^m$. If moreover $m \geq 3$ then $m + 3 \leq y^m$.*

Proof of Theorem 1.2. Write $y = y' + 1$ with $y' \geq 1$ and induct on z. For $z = 0$ the left side is 1 and the right side is $x \geq 1$. For $z = 1$ both sides equal $x \uparrow\uparrow y$. Assume $z \geq 1$ and set $m := y \uparrow\uparrow z$, so $m \geq y \geq 2$ and $y' \leq m$. By Lemma 4.3 we have $y^m \geq m+2$, so $y^m = k+1$ with $k \geq m+1$. Writing $A := x \uparrow\uparrow y = x^{(x\uparrow\uparrow y')}$ we compute

$$\begin{aligned}(x \uparrow\uparrow y) \uparrow\uparrow (z+1) = A^{(x\uparrow\uparrow y)\uparrow\uparrow z} \;&\leq\; A^{x\uparrow\uparrow m} \;=\; \left(x^{x\uparrow\uparrow y'}\right)^{x\uparrow\uparrow m} \;=\; x^{(x\uparrow\uparrow y')(x\uparrow\uparrow m)} \\ &\leq x^{(x\uparrow\uparrow m)^2} \;\leq\; x^{x\uparrow\uparrow (m+1)} \;\leq\; x^{x\uparrow\uparrow k} \;=\; x \uparrow\uparrow (k+1) \;=\; x \uparrow\uparrow \big(y \uparrow\uparrow (z+1)\big),\end{aligned}$$

using the inductive hypothesis, $x \uparrow\uparrow y' \leq x \uparrow\uparrow m$, Lemma 4.1, and $m+1 \leq k$. If $z \geq 2$ then $m \geq 4$, so $y^m \geq m+3$ by Lemma 4.3 and hence $m+1 < k$; the penultimate step is then strict. □

Remark 4.4. The argument above gives strictness whenever $y \uparrow\uparrow z \geq 3$, which covers all $z \geq 3$ and all $z \geq 2$ with $y \geq 3$. The remaining case is $y = z = 2$ with $x \geq 2$ arbitrary, and there strictness holds too:

$$(x \uparrow\uparrow 2) \uparrow\uparrow 2 = (x^x)^{x^x} = x^{x^{x+1}}, \qquad x \uparrow\uparrow (2 \uparrow\uparrow 2) = x \uparrow\uparrow 4 = x^{x^{x^x}},$$

and $x + 1 < x^x$ for $x \geq 2$. Hence the inequality of Theorem 1.2 is strict for every $z \geq 2$; only the uniform argument degrades at $y \uparrow\uparrow z = 2$.

5 The dichotomy

Proof of Corollary 1.4. Monotonicity along a root rotation is exactly (HT_r) instantiated at the values of the three subtrees; since $\mathrm{ev}_{r,a}(T) \geq 2$ for $a \geq 2$, the hypotheses of Theorems 1.1 and 1.2 are met. Non-strictness at rank 3 is Example 3.3. Strictness at rank 4 is Theorem 1.2. □

Proof of Proposition 1.3. Induction on the derivation of the cover. At the root the claim is Theorem 1.5 instantiated at the values of the three subtrees, which are all ≥ 2 because $a \geq 2$. If the rotation takes place inside the left subtree, one uses monotonicity of $\mathrm{H}_r(\cdot, v)$ in its first argument; inside the right subtree, monotonicity of $\mathrm{H}_r(u, \cdot)$ in its second argument, valid for $u \geq 2$. Both monotonicity statements hold for every $r \geq 1$ and are proved by a double induction on the rank and on the argument. The formula for the right comb is immediate from $\mathrm{H}_{r+1}(a, n+1) = \mathrm{H}_r(a, \mathrm{H}_{r+1}(a, n))$ together with $\mathrm{H}_{r+1}(a, 1) = a$. □

The mechanism is worth stating plainly. At rank 3 the identity $(x^y)^z = x^{yz}$ turns (HT_r) into $yz \leq y^z$, an inequality which is *tight*: it becomes an equality at $(y, z) = (2, 2)$, and this single coincidence is the source of every collapse in Example 3.3. At rank 4 there is no such identity, the comparison is genuinely two-dimensional, and the slack is enormous.

6 General rank

We record the general statement and the shape of a proof.

We now prove Theorem 1.5. The engine is the following inequality; write

$$\mathrm{H}_{r-1}\big(\mathrm{H}_r(x,a),\mathrm{H}_r(x,b)\big) \;\le\; \mathrm{H}_r(x,\,a+b). \tag{SUM$_r$}$$

Proposition 6.1. *Let $r \ge 3$ and $x \ge 2$.*

1. *If* (HT_{r-1}) *holds, then* (SUM_r) *holds for all $a, b \ge 1$.*
2. *If* (SUM_r) *holds and $u + v \le \mathrm{H}_{r-1}(u,v)$ for all $u, v \ge 2$, then* (HT_r) *holds for all $y \ge 2$.*

Proof. (1) Induct on a. For $a = 1$ both sides are equal:

$$\mathrm{H}_{r-1}\big(\mathrm{H}_r(x,1),\mathrm{H}_r(x,b)\big) = \mathrm{H}_{r-1}\big(x,\mathrm{H}_r(x,b)\big) = \mathrm{H}_r(x,b+1).$$

For the step, $\mathrm{H}_r(x,a+1) = \mathrm{H}_{r-1}(x,\mathrm{H}_r(x,a))$, so by (HT_{r-1})

$$\mathrm{H}_{r-1}\big(\mathrm{H}_{r-1}(x,\mathrm{H}_r(x,a)),\mathrm{H}_r(x,b)\big) \le \mathrm{H}_{r-1}\big(x,\mathrm{H}_{r-1}(\mathrm{H}_r(x,a),\mathrm{H}_r(x,b))\big),$$

and the inductive hypothesis bounds the inner term by $\mathrm{H}_r(x,a+b)$, giving $\mathrm{H}_{r-1}(x,\mathrm{H}_r(x,a+b)) = \mathrm{H}_r(x,a+b+1)$.

(2) Induct on z, with $z = 0,1$ as in Theorem 1.2. For the step set $m := \mathrm{H}_r(y,z) \ge 2$; then $\mathrm{H}_r(\mathrm{H}_r(x,y),z+1) = \mathrm{H}_{r-1}(\mathrm{H}_r(x,y),\mathrm{H}_r(\mathrm{H}_r(x,y),z)) \le \mathrm{H}_{r-1}(\mathrm{H}_r(x,y),\mathrm{H}_r(x,m)) \le \mathrm{H}_r(x,y+m) \le \mathrm{H}_r(x,\mathrm{H}_{r-1}(y,m)) = \mathrm{H}_r(x,\mathrm{H}_r(y,z+1))$. □

Proof of Theorem 1.5. Induction on r. The base case $r = 3$ is Theorem 1.1. For the step, the inductive hypothesis is exactly (HT_{r-1}), so Proposition 6.1(1) gives (SUM_r), and Proposition 6.1(2) then gives (HT_r), the hypothesis $u + v \le \mathrm{H}_{r-1}(u,v)$ for $u, v \ge 2$ being Lemma 6.2. □

Lemma 6.2. *For $u, v \ge 2$ and $s \ge 2$ we have $u + v \le \mathrm{H}_s(u,v)$.*

Proof. $u + v \le uv = \mathrm{H}_2(u,v)$, and $\mathrm{H}_2(u,v) \le \mathrm{H}_s(u,v)$ by monotonicity in the rank. □

Strictness

Lemma 6.3. *For $u, v \ge 2$ with $u \ge 3$ or $v \ge 3$, and $s \ge 2$, we have $u + v < \mathrm{H}_s(u,v)$. Equality $u + v = \mathrm{H}_s(u,v)$ occurs exactly at $u = v = 2$, where both sides are 4.*

Proof. $u + v < uv$ holds iff $(u-1)(v-1) > 1$, i.e. iff $u \ge 3$ or $v \ge 3$; and $uv = \mathrm{H}_2(u,v) \le \mathrm{H}_s(u,v)$. At $u = v = 2$ we have $u + v = 4 = \mathrm{H}_2(2,2) = \mathrm{H}_s(2,2)$ by Remark 1.7. □

Proof of Theorem 1.6. (1) $\mathrm{H}_r(\mathrm{H}_r(x,y),1) = \mathrm{H}_r(x,y) = \mathrm{H}_r(x,\mathrm{H}_r(y,1))$ since $\mathrm{H}_r(a,1) = a$. (2) is Theorem 1.1. For (3) we induct on $r \ge 4$, the base $r = 4$ being Theorem 1.2 with Remark 4.4. Let $r \ge 5$ and write $z = w + 1$ with $w \ge 1$, and $m := \mathrm{H}_r(y,w) \ge 2$.

If $y \ge 3$ or $w \ge 2$, then $y \ge 3$ or $m \ge 4$, so Lemma 6.3 makes the last step of the proof of Proposition 6.1(2) strict, and strictness propagates because $\mathrm{H}_r(x,\cdot)$ is strictly increasing.

The remaining case is $y = z = 2$. Put $A := \mathrm{H}_r(x,2) = \mathrm{H}_{r-1}(x,x) \ge 4$. Then

$$\mathrm{H}_r(\mathrm{H}_r(x,2),2) = \mathrm{H}_{r-1}(A,A), \qquad \mathrm{H}_r(x,\mathrm{H}_r(2,2)) = \mathrm{H}_r(x,4) = \mathrm{H}_{r-1}\big(x,\mathrm{H}_{r-1}(x,A)\big),$$

using $\mathrm{H}_r(2,2) = 4$ from Remark 1.7. Since $A = \mathrm{H}_{r-1}(x,x)$, the required inequality

$$\mathrm{H}_{r-1}\big(\mathrm{H}_{r-1}(x,x),A\big) < \mathrm{H}_{r-1}\big(x,\mathrm{H}_{r-1}(x,A)\big)$$

is the inductive hypothesis at rank $r-1$ applied to (x,x,A), and $A \ge 2$. □

The auxiliary facts used above — monotonicity of $\mathrm{H}_r(x,\cdot)$ in its second argument, monotonicity in the rank, and $n+1 \le \mathrm{H}_r(x,n)$ for $x \ge 2$, $r \ge 2$, $n \ge 1$ — are elementary and are proved in the formalization by a sequence of inductions.

The hypotheses of Theorem 1.5 are sharp. For $y = 1$ the inequality fails at every rank $r \ge 3$: then $\mathrm{H}_r(x,1) = x$, so the left side is $\mathrm{H}_r(x,z)$, whereas $\mathrm{H}_r(1,z) = 1$ makes the right side $\mathrm{H}_r(x,1) = x$, and $\mathrm{H}_r(x,z) > x$ as soon as $z \ge 2$. For $x = 0$ the operations degenerate.

7 The Lean development

The development is written for Lean 4 (v4.30.0) with Mathlib (v4.30.0) and is available at

`https://github.com/turahigashi/hyper-tamari`

It consists of three files, `HyperTamari/Basic.lean`, `HyperTamari/Tetration.lean` and `HyperTamari/General.lean`, together with the raw `lake build` output recording the axiom audit (`logs/axiom-audit.txt`). It contains, among others:

- `hyper`, the operation of Definition 2.1, together with `hyper_one`, `hyper_two`, `hyper_three` certifying Lemma 2.2;
- `mul_eq_pow_iff` and `ht_rank3_eq_iff`, Theorem 1.1;
- `BTree`, `eval` and the cover relation `Rot` (right rotation at an arbitrary node), with `Rot.eval_le_rank3` and `Rot.eval_le_rank4`, Proposition 1.3;
- `eval_rightComb`, the closed form for the right comb;
- `rank3_rotation_not_strict`, Example 3.3;
- `mul_le_pow_of_two_le` and `mul_lt_pow_of_three_le`, Lemma 3.1;
- `sq_tet_le`, Lemma 4.1, and `add_two_le_pow`, `add_three_le_pow`, Lemma 4.3;
- `tet_ht` and `tet_ht_strict_of_three_le`, Theorem 1.2;
- `rank3_vs_rank4_dichotomy`, Corollary 1.4;
- `sum_lemma` and `ht_general`, Proposition 6.1 and Theorem 1.5, together with the auxiliary `succ_le_hyper`, `hyper_mono_arg`, `hyper_rank_mono` and `add_le_hyper`;
- `hyper_mono_base` and `Rot.eval_le`, the general-rank form of Proposition 1.3;
- `tet_ht_strict_of_two_le`, the strengthening in Remark 4.4;
- `hyper_two_two` ($\mathrm{H}_r(2,2) = 4$) and `hyper_one_base`, Remark 1.7;
- `add_lt_hyper`, `ht_strict_general` and `equality_classification`, Theorem 1.6.

Every numbered statement of Sections 2–6 has a named counterpart in the development; the list above gives the correspondence in full.

Draft note — not part of the submission. The artifact has *not* yet been deposited in a long-term archive, so no DOI is quoted above. A version DOI will be minted and inserted here at the moment this paper is submitted, and the archive record will be linked to the submission. Until then the repository URL is the only pointer.

No declaration uses `sorry`, `native_decide` or any private axiom. Every one of the **77** theorems in these three files is audited with `#print axioms`, and the audit is machine-checked to cover them all (declared 77, audited 77, missing 0); `sorryAx` does not occur. The axiom dependencies are `propext` alone (2 theorems), `propext, Quot.sound` (45) and `propext, Classical.choice, Quot.sound` (30); the uses of choice come from library lemmas on `Nat` and are not essential to the arguments.

8 Related work

Csákány and Waldhauser [1] introduce the associative spectrum and show that exponentiation is Catalan: for pairwise distinct prime leaves, distinct bracketings give distinct values. Their concern is the number of values; the order in which those values arrange themselves is not addressed, and the Tamari lattice appears in their bibliography only as the source of the enumeration of bracketings. Huang and Lehtonen [5] extend the invariant to the *associative-commutative spectrum* and determine it for exponentiation (their Proposition 6.1.5): the associative spectrum of exponentiation is C_{n-1} and its ac-spectrum is n^{n-1}. Their question is again how many distinct term operations the bracketings produce; ours is how the resulting values are *arranged*, and neither the Tamari order nor any monotonicity statement occurs in that line of work. Huang, Mickey and Xu [6] compute the associative spectrum of a further operation. Tamari [7] enumerated the bracketings, and Friedman and Tamari [2] proved that the right-rotation order on them is a finite lattice — their title already calls the relation a *loi demi-associative*, which is precisely the law we show to run through the whole hyperoperation hierarchy. Zeilberger [8] gives a sequent calculus for that law. Gryszka [4] studies the comparison of power towers by normalizing them, which concerns right-associated towers rather than general bracketings.

Use of AI assistance

In accordance with current disclosure practice, the author states the following.

Artificial-intelligence assistants were used substantially in the preparation of this work. The question studied here was formulated by the author. Large language models were then used to search for the shape of the proofs, to write the entire Lean 4 development, to draft this text, and to check the literature; further language models were used as adversarial readers, and their criticism led to the present form of the main theorem, in particular to the classification of the equality cases in Theorem 1.6. All mathematical decisions, the choice of what to claim, and the final text are the author's responsibility.

The reader is asked to note what this does and does not mean. Every mathematical assertion in Sections 3–6 has been formalised and checked by the Lean 4 kernel, and the accompanying development is distributed with the paper; the correctness of those statements therefore does not rest on the reliability of any language model. What does rest on human and machine judgement, and is stated here so that it can be checked independently, is the choice of definitions, the claim of novelty, and the attribution of prior work.

Draft note — not part of the submission. Items marked † in the list below have *not* been consulted in the original at the time of writing; the bibliographic data were taken from secondary sources and independently cross-checked against Crossref, but the contents were not read. They are: **[FT]** Friedman–Tamari 1967, cited only for the origin of the lattice and of the term *loi demi-associative* (the phrase occurs in the title itself); **[Gry]** Gryszka 2022, cited only as related work on comparing power towers; **[Tamari]** Tamari 1962, cited only for the enumeration of bracketings and for the two combs being the extreme elements. None of the results of the present paper depends on these three items: they are cited for attribution and context, and every mathematical statement used in the proofs is either proved here or formalised in the accompanying Lean development. Before submission each of the three must be checked in the original, and this note removed by setting `\draftfalse`.

References

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